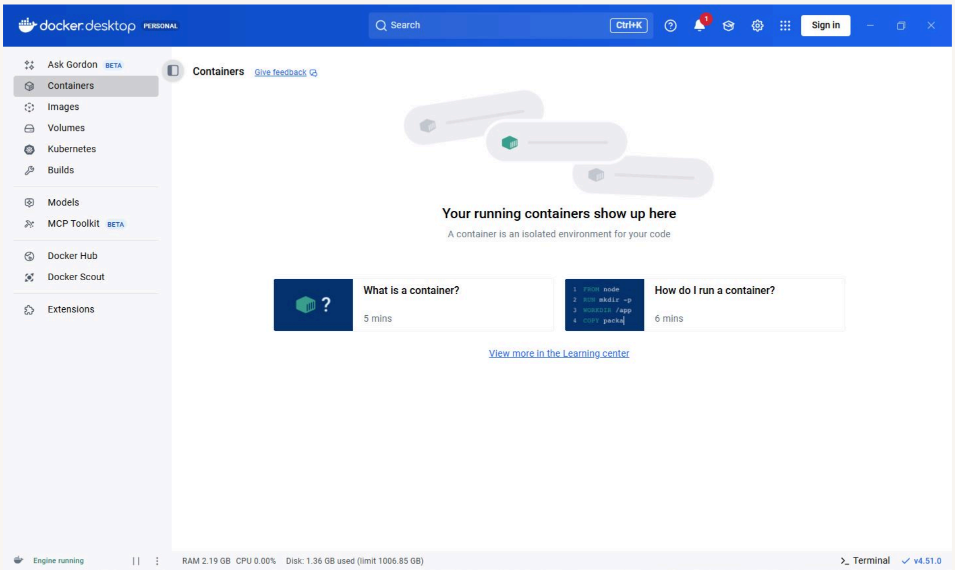
nextwork.org

Deploy an App with Docker

Mohammed Dawoud Mota



Introducing Today's Project!

What is Docker?

Docker is a platform that packages an application and its dependencies into lightweight containers. These containers run consistently across different environments, making apps easier to develop, test, deploy, and scale without compatibility issues.

One thing I didn't expect...

One thing I didn’t expect in this project was how quickly small tasks became interconnected. Solving one issue often revealed deeper improvements needed, which helped me understand the system more clearly.

This project took me...

This project took me approximately 1 hours to complete.

Understanding Containers and Docker

Containers

Containers are used to package up an application and everything needed for that application in one file. This allows developers to run the package instead of just the application. Make it work on all machines.

A container image is a blueprint or template for making containers. It tells Docker exactly what to put inside each container - things like your app's code, the libraries it needs, and any other required files.

Docker

Docker is a tool for creating and managing containers. Docker Desktop is a program that makes it easy to work with Docker. Engineers use Docker Desktop to build, test, and deploy applications right from their computers.

The Docker daemon is the background process that manages the Docker containers on your computer. It takes the command from the Docker Client and does the heavy lifting of building, running, and distributing your containers.

M
Mohammed Dawoud M...
NextWork Student

Running an Nginx Image

Nginx is a web server, which is a program you use to run websites and web apps.

The command I ran to start a new container was docker run -d -p 80:80 nginx

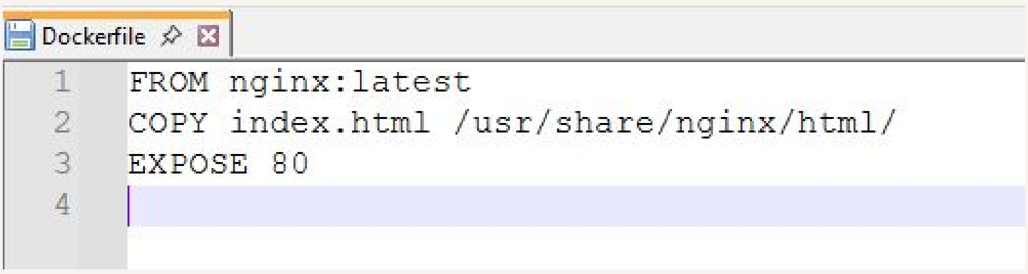
Mohammed Dawoud M...
NextWork Student

Creating a Custom Image

A Dockerfile is a document with all the instructions for building your Docker image. Docker would read a Dockerfile to understand how to set up your application's environment and which software packages it should install.

My Dockerfile tells Docker three things: the image starts as a copy of Nginx, replaces the HTML file that was there, and wants the traffic to come through port 80.

The command I used to build a custom image with my Dockerfile was docker build -t my-web-app . The '.' at the end of the command tells Docker to find the Dockerfile in the current directory



Running My Custom Image

There was an error when I ran my custom image because port 80 was being used by another container. I resolved this by stopping that container and then running the one I wanted.

The container image is the blueprint that tells Docker the application code, dependencies, libraries what should in in the container The container is the actual software that's created from this image and running the web server displaying your image

Elastic Beanstalk

AWS Elastic Beanstalk is a service that makes it easy to deploy cloud applications without worrying about the underlying infrastructure. You just upload the code, and everything else is done for you.

Deploying my custom image with Elastic Beanstalk took me around 15 minutes to deploy